

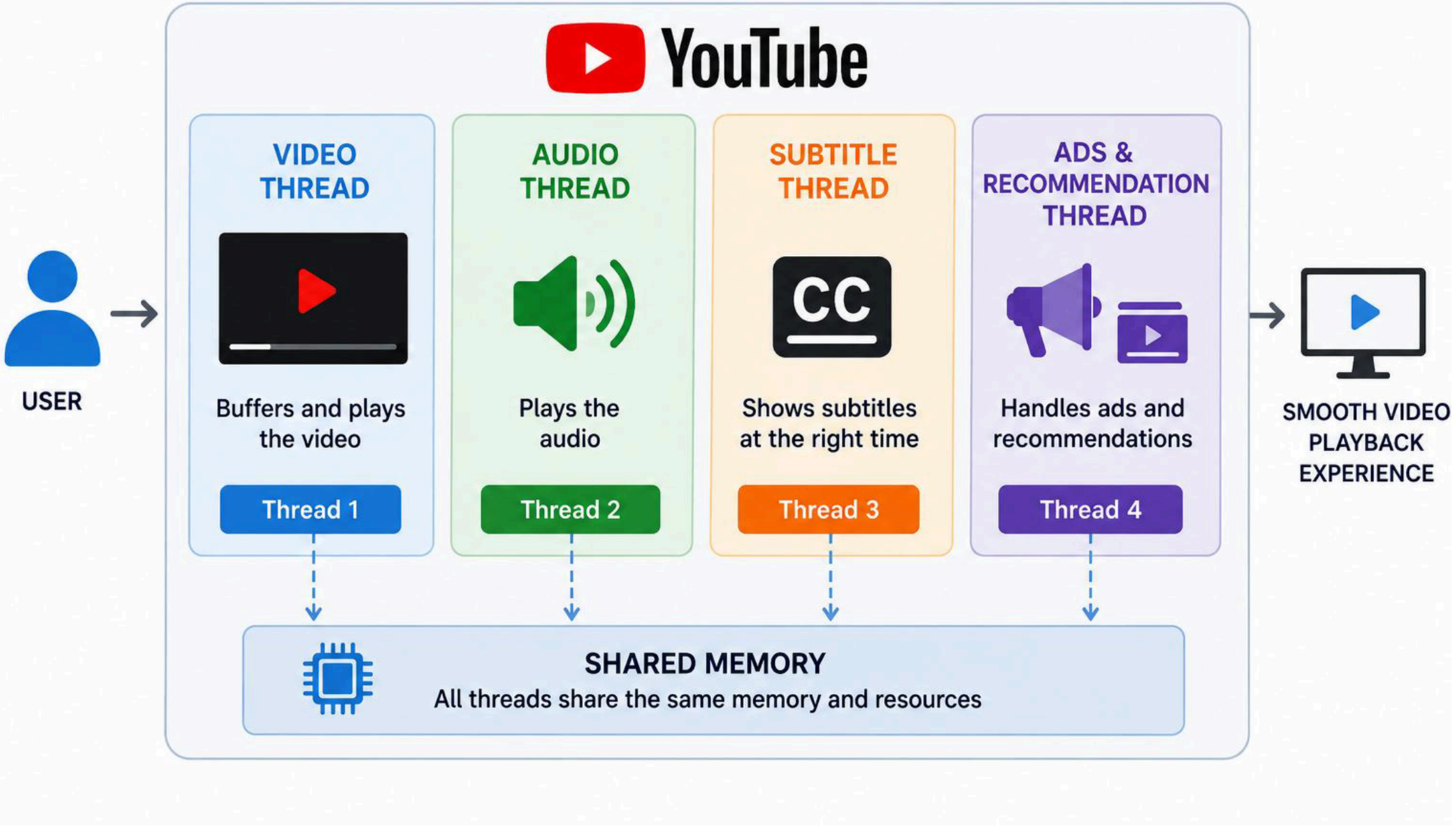


# Multithreading

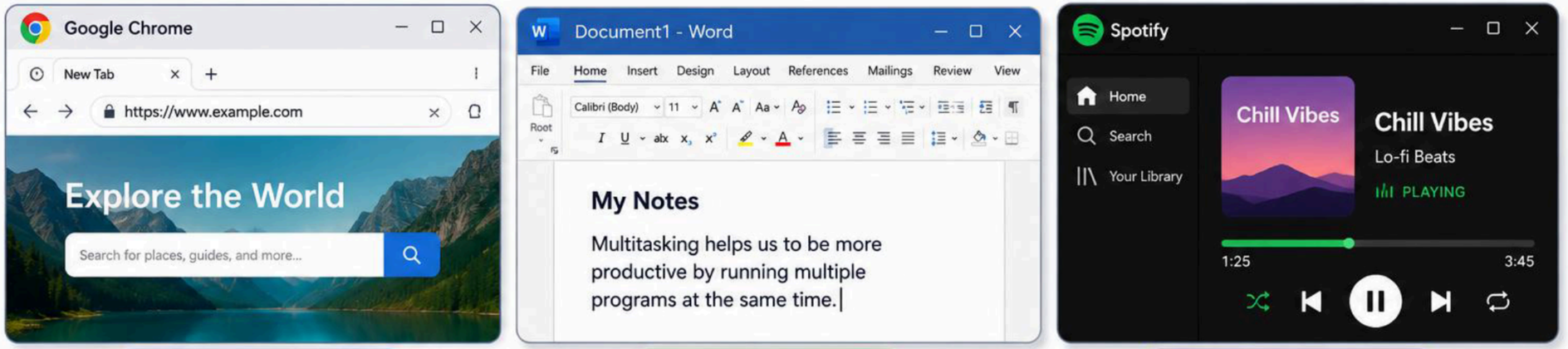
Trainer’s Name: Ms. Shanmuga Priya

# MULTITHREADING IN YOUTUBE APP

Many threads work inside one single application at the same time



# MULTITASKING




**Chrome** (Exploring)

**Word** (Typing)

**Spotify** (Music)



**EACH APP HAS ITS OWN MEMORY AREA**

**Chrome Memory Area**

This area is used only by Chrome.

**Word Memory Area**

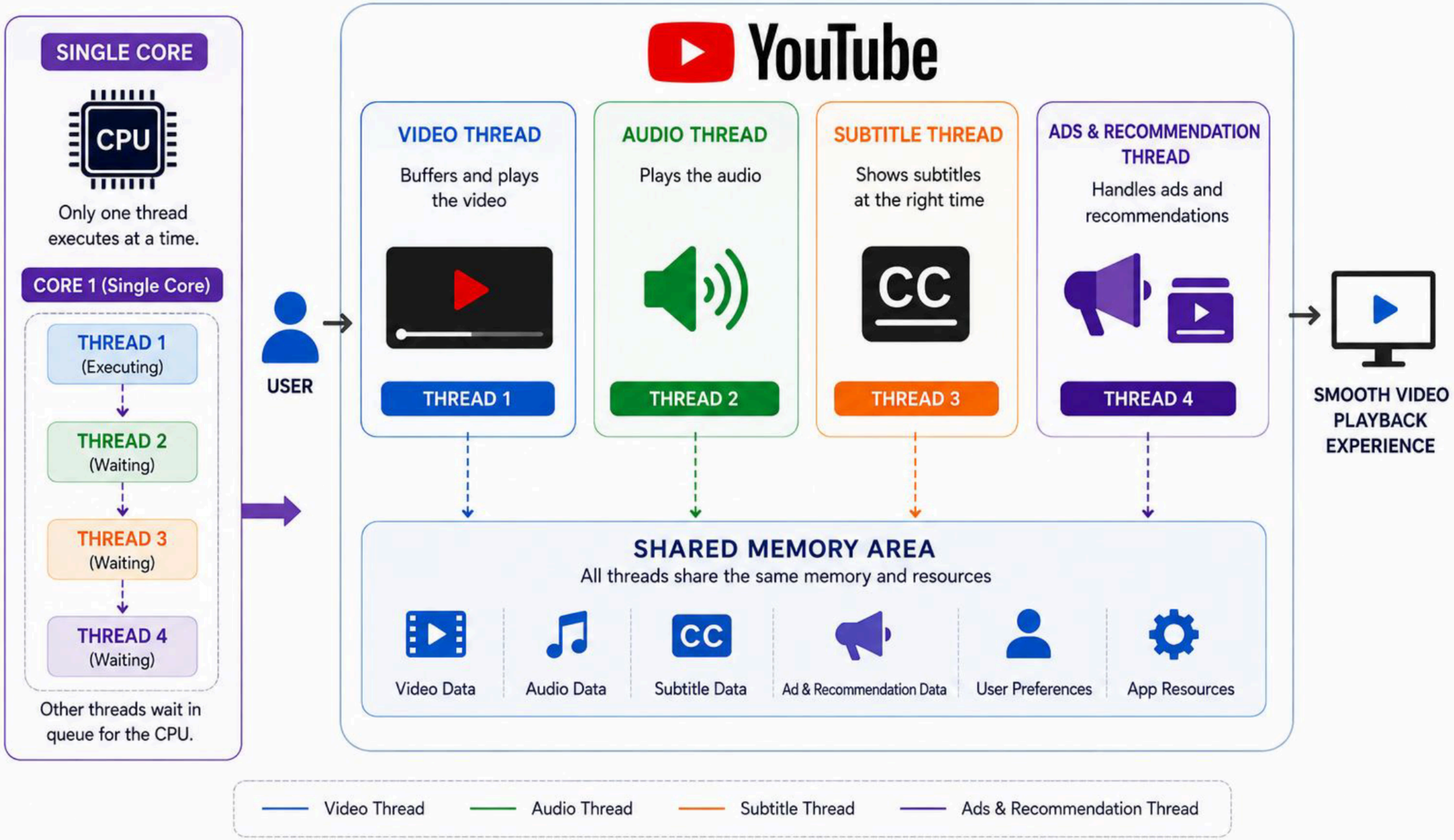
This area is used only by Word.

**Spotify Memory Area**

This area is used only by Spotify.

# MULTITHREADING WITH CONCURRENCY

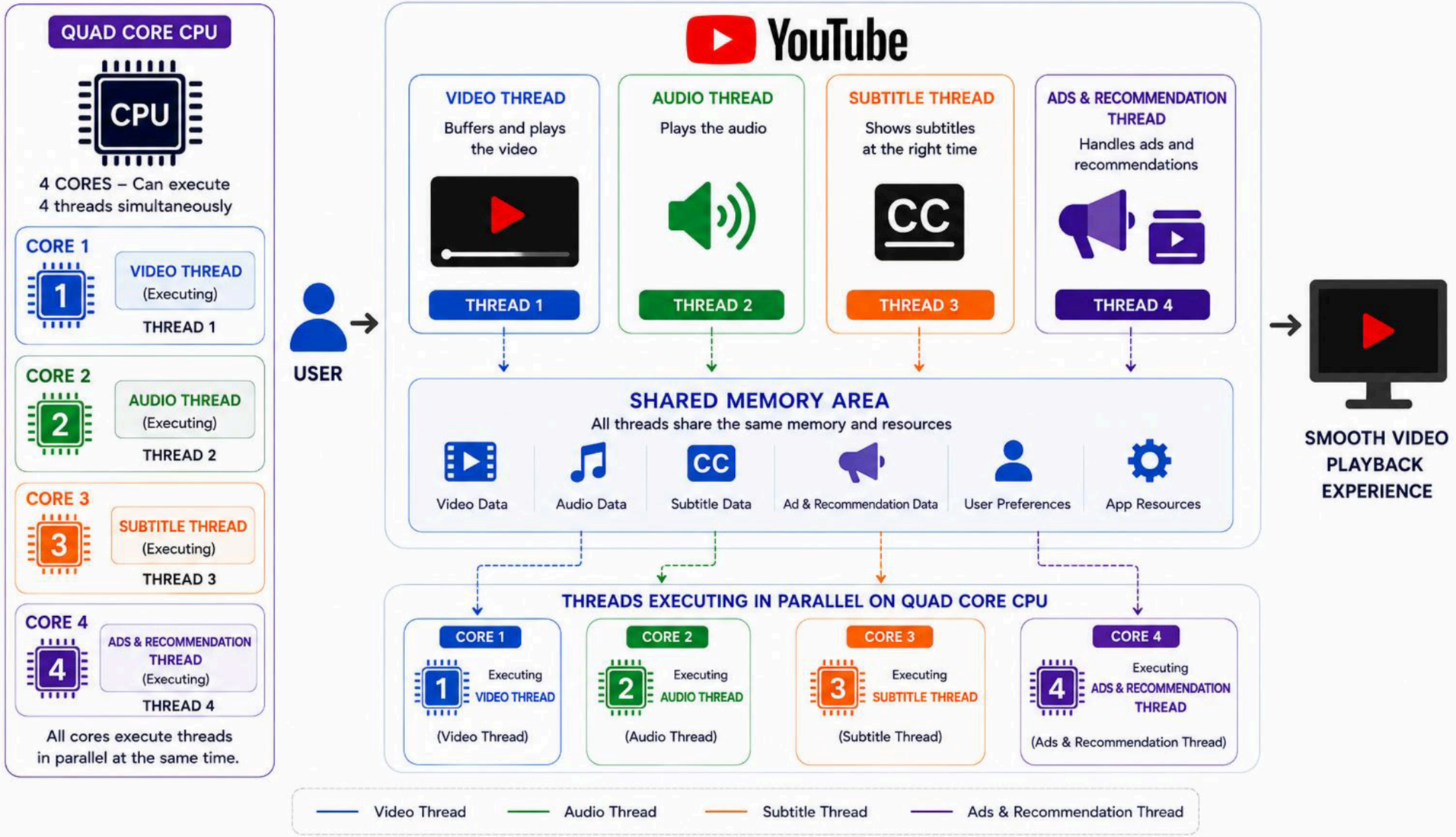
Multiple threads work inside one single program at the same time (Share the same memory area)



# MULTITHREADING WITH PARALLELISM

Multiple threads work inside one single program at the same time (Share the same memory area)

With Quad Core CPU – Threads can run truly in parallel on different cores



## Ways to create thread

1) Extending Thread class
2) Implementing Runnable interface

## sleep method

```
//1000 ms - 1 sec
try {
    Thread.sleep(5000); // wait for 5 seconds
}
catch (InterruptedException e)
{
    System.out.println("Alarm was interrupted!");
}
    System.out.println("Ring Ring! Wake up!");
}
}
```

# join method

```
//thread1 is Cutting Vegetables
CookingTask cuttingThread = new CookingTask("Cutting Vegetables");//thread1 object
//thread2 is Boiling Vegetables
CookingTask boilingThread = new CookingTask("Boiling Vegetables");//thread2 object

cuttingThread.start();


// main thread waits until cuttingThread completes its process
try {
cuttingThread.join();
} catch (InterruptedException e) {
e.printStackTrace();
}

boilingThread.start(); // start boiling only after cutting
```

# THANK YOU